

Laboratory 2: PID Controllers

Preparation:

- Read the instructions carefully and make sure you are prepared (for example, theory wise) to complete the lab.
- **Note: Results obtained in Laboratory 2 will be used in Laboratory 3.** So, please take note of the number of the tank apparatus, and use the same tank for Laboratory 3.
- We recommend that you revise the following concepts: open- and closed-loop control, block diagram, input, output, reference, control signal.
- Download the file `laboration2.zip` from the course's Canvas room, unzip and save the files in your work folder.
- Open the model `laboration2.slx` in MATLAB and verify that it looks like the one in Figure 2

1 Introduction

The purpose of this laboratory is to start learning in practice how to use a PID-controller, the most common type of controller in the process industry. The PID-controller has a simple structure, which gives us the opportunity to try and understand fundamental concepts and principles of automatic control.

Laboratory setup

The double-tank process (Figure 1a) used in this lab consists of two water tanks connected in series. Such tanks can be used for example as a buffer between two sections in a chemical plant in order to make material flowrates between sections more constant. Furthermore, tanks are used in the process industry to mix different products and to control pressure, concentration pH and temperatures, where common control objectives are to keep the tank level constant while the material flows vary; or to keep the material flows constant while the tank level varies. The same concept can be extended to data communication (cache), logistics (product storage) or even finance (bank account).

2 Fundamental concepts in automatic control

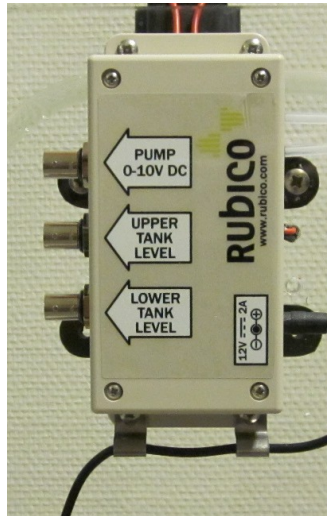
Why use automatic control?

In most cases, automatic control is used to make a process behave in a certain (desired) way. Some examples are to make a process more effective by reducing energy consumption or increasing product quality; or to make an unstable processes stable, thus reducing the risks of accidents.

Therefore, desired behaviour can be different depending on the process. When it comes to the double-tank process in this lab, some examples of control goals are:



(a) Double-tank process.



(b) Connector block from the tank



(c) Connector block to the computer.

Figure 1: Lab equipment.

- Keep a tank level constant, or in automatic control lingo, to keep the measured variable (or output) at its reference.
- Reference changes between tank levels are expected to happen fast, without tanks overflowing or oscillatory responses.
- The control system should be able to mitigate external disturbances, for example when someone fills a tank with a cup of water, or when someone opens the valve in the upper tank.
- It is desired to minimize pump wear caused by too fast or too aggressive variations in the flowrate.

The aforementioned properties are generic and applicable to many different contexts. Usually, it is also desired to quantify these control goals by specifying for example how many centimeters consists a too large oscillation.

Exploring the process

Exercise 2.1 *Get familiar with the lab equipment. Check how disturbances are introduced in the process. Is there measurement noise? If so, is it possible to remove it? Identify the process, controller, manipulated variable and measured variables in Figure 3*



Exercise 2.2 *In order to run the system in closed-loop (PID designed in the computer and double-tank process) one needs to first physically connect the two components. This is done by using cables to connect the process' connection box and the computer's connection box. Figure 2 summarizes which inputs in*

Figure 1c are used. AI and AO mean Analog Input and Analog Output. The levels of the upper and lower tanks are input signals to the controller and the output of the controller is the pump's voltage.

Observe that sometimes one or more contacts in the computer's coupling box can malfunction. In this case, one can change which contacts are being used by double-clicking the double-tank figure in Simulink and changing the port numbers.

Connect the double-tank process to the computer.

Exercise 2.3 Extract the file laboration2.zip. Then open MATLAB/Simulink in your computer and open the file Lab2R0004E.slx which is the Simulink model for the experiment. This Simulink diagram should look like Figure 2. Double-click on the Scope block to check the manipulated variable and measured variable during the simulation. Make sure that the controller is set to "Manual" mode, and the measured variable being used is the level of the upper tank.

Change the value of "Manuell Styrsignal" to 0.5 and run the model (click on the green Play button). If the connections are working correctly, the pump should start working and the water level in the upper tank should increase. Make sure that the disturbance valve in the upper tank is closed.

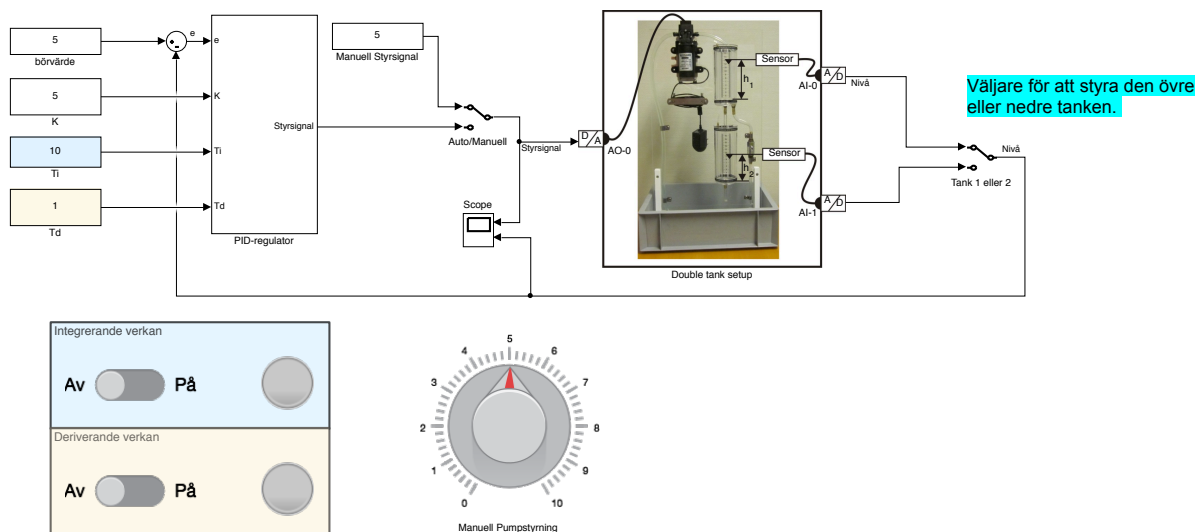


Figure 2: Controller in Simulink.

Block diagram

Block diagrams are frequently used in automatic control to represent the relationship between processes and variables of interest for a controller design task. All superfluous components should be simplified or eliminated. It is important that you understand how to create a block diagram as a representation of the control problem you want to solve. In this lab, we are interested in studying the (dynamic) relationships between reference variable, measured variable and manipulated variable.

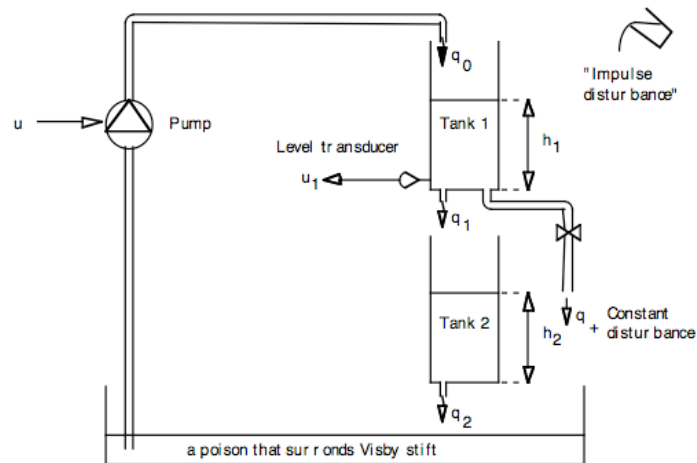


Figure 3: Double-tank process.

Exercise 2.4 Sketch a block diagram for the double-tank + computer process, where the tank level is controlled by the pump voltage. Indicate in the block diagram the process and the measured variables. Convince yourself that you can understand the relationship between the block diagram you just sketched and the real process you want to control.



Open-loop control

It is mandatory to understand the difference between open-loop and closed-loop control. In practice, many control solutions are based on a combination of the two strategies. The biggest difference between them is that in open-loop control, the control signal is not calculated as a function of the measured variable, but rather taken from a data table or operation data.

Your objective now is to use open-loop control to stabilize the level in the tanks. In order to do so, you will create a static model based on process operation data, or in other words, find the equivalence between input (pump voltage) and output (tank level).

Exercise 2.5 Using the Simulink model, set the pump voltage at 0.5V, wait until the level of the upper

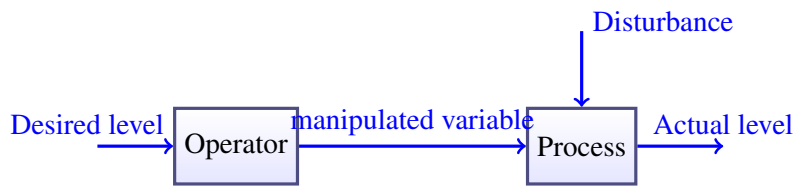
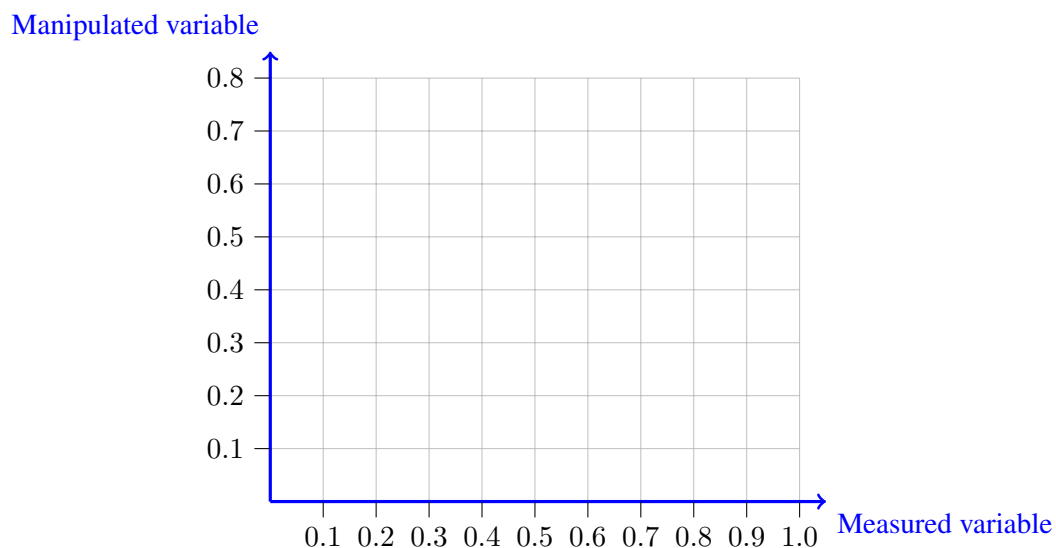


Figure 4: Open-loop control.

Manipulated variable	Measured variable
0.4	
0.5	
0.6	

tank stabilizes and register its value ¹ in the Table below. Repeat the experiment for the other values. Plot the relationship between measured variable and manipulated variable.



Exercise 2.6 Using the data table in from the previous Exercise, set the control signal that should drive the upper tank level to 0.5. Then, using linear extrapolation, try to find the control signal that drives the upper tank level to 0.7. Remember that in this experiment, you are not allowed to correct the control signal based on what you observe in the measured variable. What happens if you open the disturbance valve?

¹You can read the value of a signal by right-clicking on a line in the Simulink model and choosing Show Value Label of Selected Port

Closed-loop control

Now, before going into PIDs, you will try to control the process manually by acting as the controller. You are free to look at the process and use information from the measured variable to correct your control signal. See the equivalent block diagram in Figure 5.

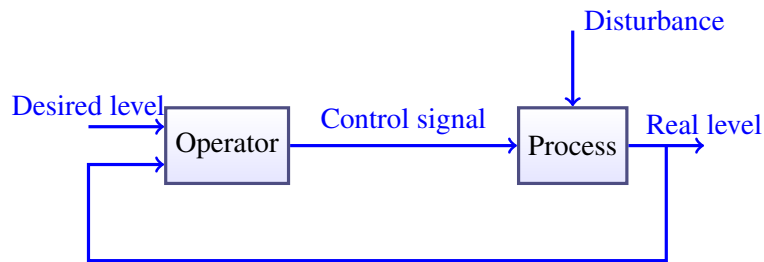


Figure 5: Closed-loop control

Exercise 2.7 Try one more time to change the upper tank level from 0.5 to 0.7, but this time you can correct the control signal based on the current value of the upper tank level. Compare the results to Exercise 2.5. What limits how fast you can change the tank level? Explain.

Try to keep the upper tank level constant when you classmate introduces disturbances in the system (by opening the upper tank's valve). Does open-loop or closed-loop control work better in this case? Why?

Comparing the upper and lower tanks

We are now going to investigate how the task of controlling the level of the upper tank differs from controlling the level of the lower tank.

Exercise 2.8 In your Simulink model, change the measured variable to the level in the lower tank and repeat Exercise 2.7. What limits how fast the level of the lower tank can be changed?

Which tank is easier to control? Why?

3 PID control

It is finally time to start using controllers to regulate the tank levels in the double-tank process. In general, a controller is an algorithm that compares the reference value and the measured variable, and calculates an appropriate control signal based on that difference.

P-controller

First we will try the P-(proportional) controller. In the Simulink model, make sure the control signal switch is on Auto and that Integral and Derivative components are disabled.

In a P-controller, the control signal u is calculated as follows:

$$e(t) = r(t) - y(t) \quad (1)$$

$$u(t) = Ke(t) \quad (2)$$

where r is the reference signal, and y is the measured signal. The difference e between reference and measured variable is called control error and u is the calculated control signal. In the case of P-control, the control signal is proportional to the control error, and the proportionality constant K is usually called controller gain.

Exercise 3.1 We will explore now how the properties of the controller (how fast the error converges, how aggressive it is, etc) vary as a function of the controller gain K . Before the start of every experiment, use the upper tank level as measured variable, set the reference value to $r = 0.5$, and make sure the disturbance valve is closed.

Set $K = 5$ and change the reference value of the upper tank level from $r = 0.5$ to $r = 0.7$. Wait until the upper tank level is stable. Are there any differences between positive and negative changes in the reference value (from $r = 0.5$ to $r = 0.7$ and from $r = 0.7$ to $r = 0.5$).

Repeat the experiment for $K = 1$ and $K = 10$. How does the control error and how fast the closed-loop system response vary with different values of K ?

Increase the gain to $K = 20$ and repeat the experiment above. What are the main differences compared to the results from $K = 10$?

Exercise 3.2 The objective now is to investigate how the P-controller performs under disturbances. You can generate both step disturbances (open the valve close to the upper tank) and impulse disturbances (fill the upper tank with a cup of water). How does the closed-loop behaviour changes with different values of K (test at least 2 different values)?

Observation! The pump capacity might not be enough to handle a fully open valve.

Exercise 3.3 Before we move on, the last task is to investigate how the controller gain affects measurement noise. Carefully observe how the measured variable is affected by different values of K . Based on your observations, in all the experiments performed in this Section, what is the best value for K ?

Exercise 3.4 You should have by now a good intuition on how K affects the closed-loop system for the upper tank process. Repeat the experiments in Exercises 3.1 - 3.3, but use the level of the lower tank as measured variable. Repeat only the experiments you deem necessary to obtain the best value of K for closed-loop control of the lower tank.

PI-controller

As you probably observed in the previous experiments, P-controllers lead to offset (a constant control error after the closed-loop system has stabilized) because an infinite gain would be needed to remove a non-zero control error. This shortcoming is addressed by the I- (integral) component of the PI-controller. In practice, the control signal is increased if the control error is positive, and decreased if the control error is negative. The control signal in a PI-controller is calculated as:

$$u(t) = K \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau \right) \quad (3)$$

In the above equation, the control signal (the voltage of the pump) is a sum of two terms: one that is proportional to the control error, and another that is the integral of the control error multiplied by a constant $\frac{K}{T_i}$.

Exercise 3.5 Investigate how the Integral gain T_i affects the level control of the upper tank. In the Simulink model, activate the Integral term, keep K constant (you can use the optimal value you obtained previously), and vary T_i between 20 and 1 (in decrements of 4, taking the values (20, 16, 12, 8, 4, 1)). How does T_i affect the closed-loop response of the upper tank, considering set-point (reference) changes and disturbances?

Exercise 3.6 Try now some different values of K and T_i and try to find the combination that gives the best closed-loop performance, considering set-point changes, disturbance rejection and even measurement noise. Which pros and cons do you observe compared to a P-controller?

Exercise 3.7 Repeat the previous experiments (the ones you deem necessary) to obtain the best values for K och T_i when controlling the level of the lower tank.

PID-controller

In the PI-controller, the control signal is calculated based on the current error and the past error values. It would be interesting to even consider future error values to anticipate and correct deviations in the control error. Although it is not possible to use future values directly (since they don't exist yet), the derivative of the error can be used to estimate future errors as follows:

$$e(t + T_d) \approx e(t) + T_d \frac{de(t)}{dt} \quad (4)$$

For small values of T_d , the approximation of the future error should be good enough to improve control performance. Therefore, the control action by a PID-controller is calculated as:

$$u(t) = K \left(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right) \quad (5)$$

which corresponds to the control action of a PI-controller plus a third term that is proportional to the derivative of the control error. The constant T_d , also called derivative time, is an indicator of how long in the future the error is being predicted.

Exercise 3.8 Investigate how T_d affects closed-loop performance of the upper tank level. In the Simulink model, do not forget to activate the Proportional-component. Start with the optimal values of K and T_i obtained in previous Exercises. Did T_d improve control performance? What are the main differences?

Exercise 3.9 Now investigate how T_d affects the control of the lower tank. Start with K and T_i from the Exercise 3.7 and varying T_d between 0.2 and 5 (in increments of 1, taking the values (0.2, 1, 2, 3, 4, 5)). Comment your results.

4 Conclusions

Exercise 4.1 List important differences between open-loop and closed-loop control.

Exercise 4.2 Discuss pros and cons of P-, PI-, and PID-controllers for the upper and lower tanks. Tip: base your discussion around the performance in set-point tracking and disturbance attenuation; how aggressive the control signal is; and what does the offset look like.

In particular, think about the differences between large and small values of K , T_i and T_d . Are there any differences in the trends comparing the upper and the lower tanks?

5 Controller settings

Complete the Table below and use the parameter values for Laboratory 3.

	Upper tank	Lower tank
P	$K =$	$K =$
PI	$K =$ $T_i =$	$K =$ $T_i =$
PID	$K =$ $T_i =$ $T_d =$	$K =$ $T_i =$ $T_d =$

6 Evaluation

6.1 Oral session

The Oral session is an opportunity for you and your group to discuss the lab instructions with the lab assistants, and also an opportunity for us to make sure you have prepared to the lab. Check the Canvas room for information on how to book your session. The goal is not to evaluate your performance in the lab (which is done in a report), but rather to check that you know what you are going to do, how it is relevant to the course's theory and what is expected from you. In other words, this is a formative evaluation activity that should be more useful to you than to us.

6.2 Report

Some points to guide you to write the report:

- Write a report that include all the answers to the questions in this lab instruction.
- Whenever relevant, provide explanation and analysis to your answers.
- Include the plots or at least the sketches of the control signal and the response of each tank (upper tank and lower tank), with each controller (P, PI and PID) for the best configuration of the controller parameters that you recorded in Section 5.
- Discuss your results, what you have learned and how you think they will be/are relevant to the course.
- The report should be submitted in Canvas before the deadline.
- Grade scale: < 80 - U, ≥ 80 - G.

See the Canvas room for more information.

7 Review history

When	Who	What
2025-04-28	DSL	Minor editing and remove grading criteria.
2024-06-10	ASY	Instructions translated to English and minor improvements.
2024-03-22	ASY	New grading criteria
2023-06-07	ASY	New grading criteria.
2023-05-26	EKoJE	New Exercise to explain how to connect the tank system to the computer.
2023-03-06	ASY	Minor improvements.